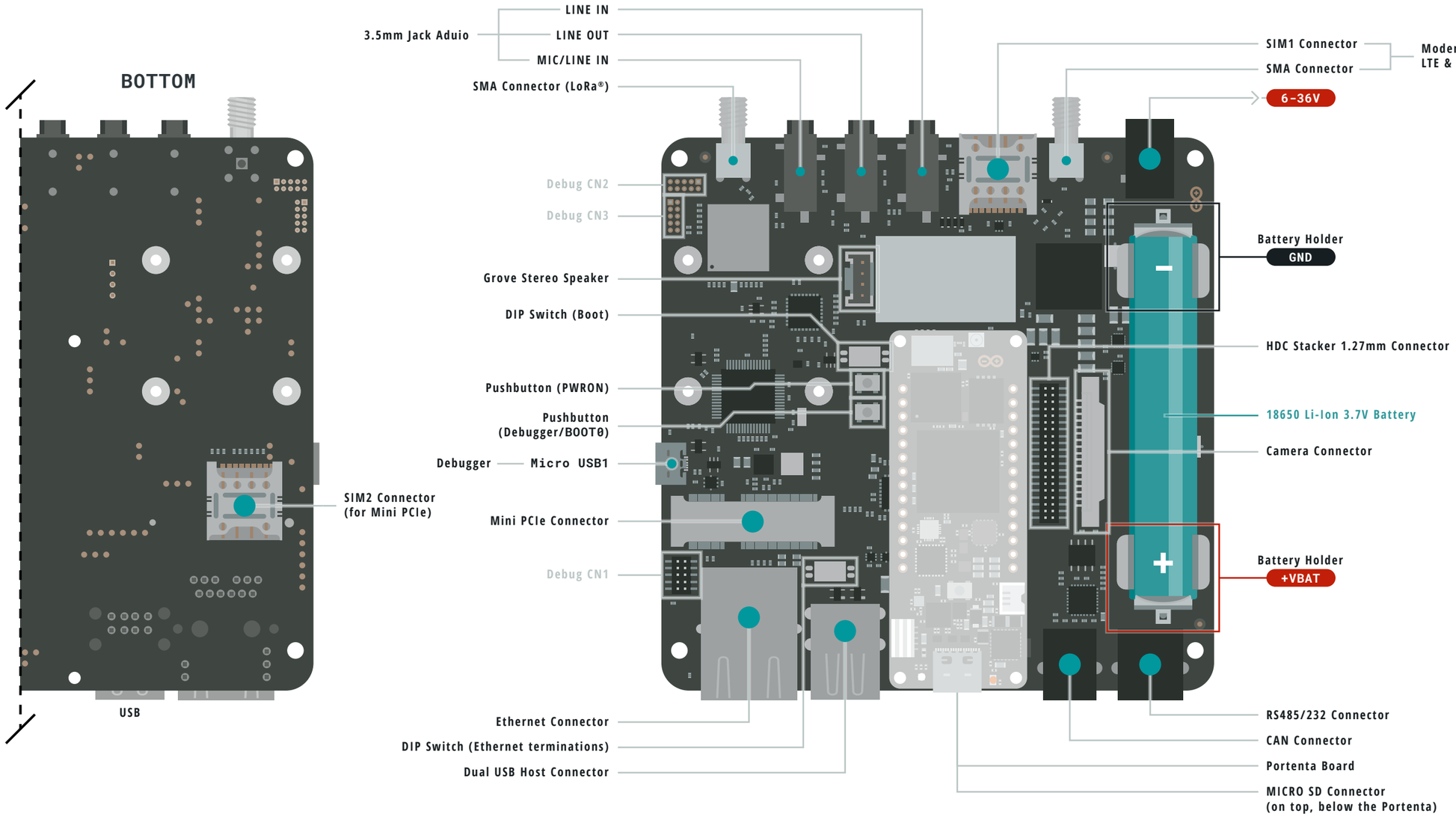


- Ground
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- Internal Pin
- SWD Pin
- Digital Pin
- Analog Pin
- Other Pin
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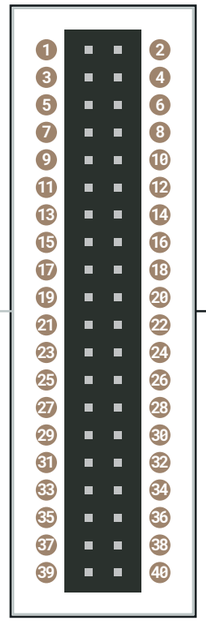
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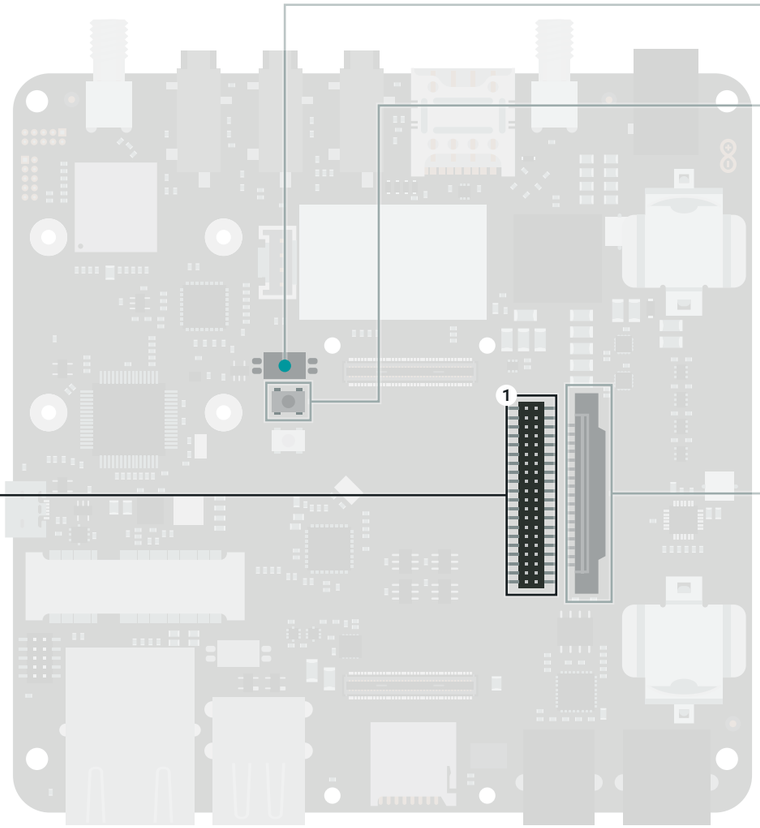
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J5 HDC Stacker Connector

Device Name (Linux)	GPIO (Linux)		
		+3V3	1
		GND	2
		J2-71 ANALOG_VREF_P	3
		J2-72 ANALOG_VREF_N	4
/sys/bus/iio/devices/iio:device0/in_voltage0_raw	167	J2-73 ANALOG_A0	5
/sys/bus/iio/devices/iio:device0/in_voltage1_raw	168	J2-75 ANALOG_A1	6
/sys/bus/iio/devices/iio:device0/in_voltage2_raw	169	J2-77 ANALOG_A2	7
/sys/bus/iio/devices/iio:device0/in_voltage3_raw	170	J2-79 ANALOG_A3	8
/sys/bus/iio/devices/iio:device0/in_voltage4_raw	171	J2-74 ANALOG_A4	9
/sys/bus/iio/devices/iio:device0/in_voltage5_raw	172	J2-76 ANALOG_A5	10
/sys/bus/iio/devices/iio:device0/in_voltage6_raw	173	J2-78 ANALOG_A6	11
/sys/bus/iio/devices/iio:device0/in_voltage7_raw	174	J2-80 ANALOG_A7	12
		GND	13
		GND	14
	107	J2-49 SAI CK	15
	137	J2-36 SPI1 CS	16
	106	J2-51 SAI FS	17
	134	J2-38 SPI1 CK	18
	108	J2-53 SAI D0	19
	136	J2-40 SPI1 CIPO	20
	109	J2-55 SAI D1	21
	135	J2-42 SPI1 COPI	22
	178	J1-74 SPDIF_TX	23
/sys/bus/iio/devices/iio:device0/in_voltage1_raw	168	J2-25 SERIAL3_TX	24
	178	J1-76 SPDIF_RX	25
/dev/ttymxc3	156	J2-27 SERIAL3_RX	26
/dev/i2c-1	145	J1-43 I2C1 SDA	27
		J2-29 SERIAL3_RTS	28
/dev/i2c-1	144	J1-45 I2C1 SCL	29
		J2-31 SERIAL3_CTS	30
		+3V8	31
/dev/i2c-3	149	J2-45 I2C2 SDA	32
/sys/class/pwm/pwmchip0/pwm6	189	J2-62 PWM6	33
/dev/i2c-3	148	J2-47 I2C2 SCL	34
/sys/class/pwm/pwmchip0/pwm7	190	J2-64 PWM7	35
		+5V	36
/sys/class/pwm/pwmchip0/pwm8	191	J2-66 PWM8	37
can1 (network)	177	J1-50 CAN0_TX	38
/sys/class/pwm/pwmchip0/pwm9	192	J2-68 PWM9	39
can1 (network)	178	J1-52 CAN0_RX	40





DIP Switch
SW1 218-2LPSTF

1	BOOT_1	J2-01 BOOT
2	BT_SEL_1	J2-03 BT_SEL

PB1 Pushbutton

6	NetPB_1	J2-05 PWRON

Camera Connector
J4 1734248

1	CSI_D3_N	J2-02 CSI_D3_N
2	CSI_D3_P	J2-04 CSI_D3_P
3	HS	J2-22 CAM_HS
4	CSI_D2_N	J2-08 CSI_D2_N
5	CSI_D2_P	J2-06 CSI_D2_P
6	GND	
7	CSI_D0_N	J2-16 CSI_D0_N
8	CSI_D0_P	J2-14 CSI_D0_P
9	GND	
10	CSI_D1_N	J2-12 CSI_D1_N
11	CSI_D1_P	J2-10 CSI_D1_P
12	GND	
13	CSI_CK_P	J2-18 CSI_CK_P
14	CSI_CK_N	J2-20 CSI_CK_N
15	GND	
16	CK	J1-66 PDM CK
17	D0	J1-68 PDM D0
18	SCL	J1-46 I2C0 SCL
19	SDA	J1-44 I2C0 SDA
20	+3V3	

TOP VIEW

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Pins 2 and 3 of J14 SJ-3524-SMT-TR are both connected to pin 20 MICBIAS of U2 CS42L52-CNZ

Audio U2 CS42L52-CNZ	LINE IN J14 SJ-3524-SMT-TR
23 AIN3A/20 MICBIAS	3
24 AIN3B/20 MICBIAS	2

Audio U2 CS42L52-CNZ	LINE OUT J12 SJ-3524-SMT-TR
14 HP/LINE_OUTB	3
15 HP/LINE_OUTA	2
31 SPKR/HP	4

Audio U2 CS42L52-CNZ	MIC/LINE IN J13 SJ-3524-SMT-TR
25 AIN2A	3
26 AIN2B	2

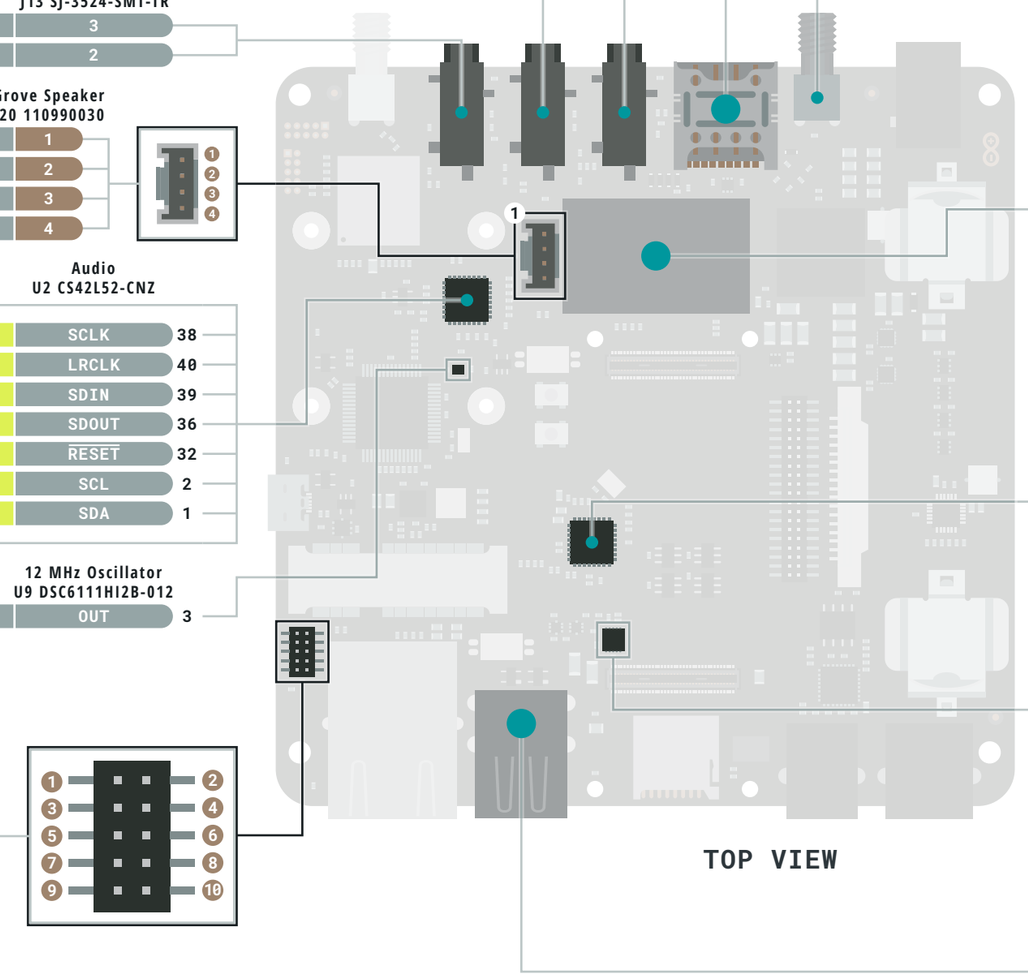
Audio U2 CS42L52-CNZ	Grove Speaker J20 110990030
4 SPK_OUTA+	1
6 SPK_OUTA-	2
7 SPK_OUTB+	3
9 SPK_OUTB-	4

Device Name (Linux)	GPIO (Linux)	Audio U2 CS42L52-CNZ		
178	J1-56 CK	SCLK	38	
178	J1-58 WS	LRCLK	40	
178	J1-62 SDO	SDIN	39	
178	J1-60 SDI	SDOUT	36	
161	J1-48 GIPO1	RESET	32	
/dev/i2c-2	J1-46 I2C0 SCL	SCL	2	
/dev/i2c-2	J1-44 I2C0 SDA	SDA	1	

Audio U2 CS42L52-CNZ	12 MHz Oscillator U9 DSC6111HI2B-012	
37 MCLK	OUT	3

Debug CN1

+3V3	1
JTAG_SWD	2
GND	3
JTAG_SCK	4
GND	5
JTAG_SWO	6
NC	7
JTAG_TDI	8
+3V3 DBG	9
JTAG_RST	10



SIM1 Connector SIM8060-6-0-14-00-A		Cellular Module U1 SARA-R412M-02B
C1	VCC	41 VSIM
C7	I/O	39 SIM_IO
C3	CLK	38 SIM_CLK
C2	RST	40 SIM_RST
C5	SW	42 SIM_DET

	J3 73100-0114	Cellular Module U1 SARA-R412M-02B
1	SMA Connector	56 ANT

Modem Cellular Module U1 SARA-R412M-02B

		GPIO (Linux)	Device Name (Linux)
12	TXD	J1-33 SERIAL1_TX	128	/dev/ttymxc1
13	RXD	J1-35 SERIAL1_RX	127	
10	RTS	J1-37 SERIAL1_RTS	126	
11	CTS	J1-39 SERIAL1_CTS	125	
18	RESET	J2-50 GIPO2	162	
15	PWR_ON	J2-54 GPIO4	164	

		USB Hub - U3 USB2514B
62	ANT DET	
17	V_USB_DET	18 BC_EN3
28	USB_D-	7 USBDP_DN3
29	USB_D+	6 USBDM_DN3

USB Hub U3 USB2514B		
31	USBDP_UP	J1-26 USB0_D_P
30	USBDM_UP	J1-28 USB0_D_N
24	SCL	J1-46 I2C0 SCL
22	SDA	J1-44 I2C0 SDA

USB Host U7 NCP383LMUAJAATXG		USB Hub U3 USB2514B
2	+5V	
3	+5V	
4	EN1	12 BC_EN1
5	EN2	16 BC_EN2
10	FLAG1	13 OCS_N1
6	FLAG2	17 OCS_N2

Dual USB Host Connector J15 61400826021		USB Host U7 NCP383LMUAJAATXG	USB Hub U3 USB2514B
1	VBUS	OUT1	
2	D-		1 USBDM_DN1
3	D+		2 USBDP_DN1
4	GND		
5	VBUS	OUT2	
6	D-		3 USBDM_DN2
7	D+		4 USBDP_DN2
8	GND		

Modem pins TXD, RXD, RTS, CTS, RESET and PWR_ON are translated from 1V8 (Modem) to 3V3 (Portenta)

Ground
Power
LED
Internal Pin
SWD Pin

Digital Pin
Analog Pin
Other Pin
High Density Connector

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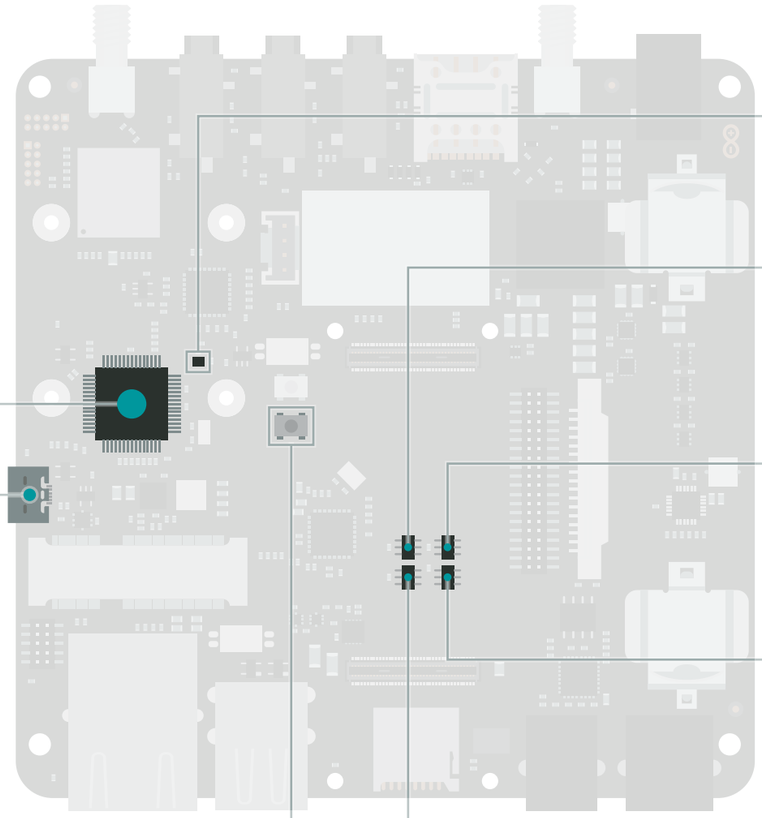
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TOP VIEW



Device Name (Linux)	GPIO (Linux)		DEBUG U15 STM32F405RGT6	
/dev/i2c-3	149	J2-45 I2C2 SDA	PA8 I2C3 SDA	41
/dev/ttymxc2 (Linux Console)	154	J2-28 SERIAL2_RX	PA9 UART TX	42
	155	J2-26 SERIAL2_TX	PA10 UART RX	43
			PA13 DBG SWDIO	46
			PA14 DBG SWCLK	49
		J1-75 JTAG SWD	PC0 JTAG SWD	8
		J1-79 JTAG SWO	PC2 JTAG SWO	10
	178	J1-78 JTAG TDI	PC3 JTAG TDI	11
		J1-73 JTAG RESET	PC5 JTAG RST	25
		J1-77 JTAG SCK	PC8 JTAG SCK	39
/dev/i2c-3	148	J2-47 I2C2 SCL	PC9 I2C3 SCL	40
			PB2 BOOT1	28
can0 (network)	178	J1-51 CAN1 RX	PB5 CAN2 RX	57
	177	J1-49 CAN1 TX	PB6 CAN2 TX	58
/dev/i2c-1	145	J1-43 I2C1 SDA	PB7 I2C1 SDA	59
	144	J1-45 I2C1 SCL	PB8 I2C1 SCL	61
/dev/i2c-2		J1-46 I2C0 SCL	PB10 I2C2 SCL	29
/dev/i2c-2		J1-44 I2C0 SDA	PB11 I2C2 SDA	30
			PB13 +5V DBG	34
		J1-79 JTAG SWO	PD2 JTAG SWO	54
			RST	7
			VBAT 3V3_DBG	1
			VDD 3V3_DBG	19
			VDD 3V3_DBG	32
			VDD 3V3_DBG	48
			VDD 3V3_DBG	64
			VDDA 3V3_DBG	13

32.768 kHz Crystal
Y3 Q13FC1350000400

2	PC14 OSC32 IN	3
1	PC15 OSC32 OUT	4

DEBUG U15 STM32F405RGT6

35 PB14 USB1_D_N	USB D-	2
36 PB15 USB1_D_P	USB D+	3

J10 Micro USB

+5V_DBG		1
	GND	5

The debugger can be powered by the 3V3 on board or by J10 Micro USB Connector.

U9 VDD
3V3_CLK

3V3_DBG
3V3

Double Diode Package
D11 BAV70M,315

12 MHz Oscillator
U9 DSC6111HI2B-012

			DEBUG U15 STM32F405RGT6
3	OUT		5 PH0
4	VDD 3V3_CLK		

U16 74LVC1G157GW

			DEBUG U15 STM32F405RGT6
3	I0	J1-36 SERIAL0_RX	
1	I1	J1-35 SERIAL1_RX	
4	Y		52 PC11 UART3_SNIFF
6	SEL		20 PA4 UART_CH1-2
5	VCC 3V3_DBG		

U17 74LVC1G157GW

			DEBUG U15 STM32F405RGT6
3	I0	J2-25 SERIAL3_TX	
1	I1		
4	Y		15 PA1 UART4_SNIFF
6	SEL		21 PA5 UART_CH3-4
5	VCC 3V3_DBG		

U18 74LVC1G157GW

			DEBUG U15 STM32F405RGT6
3	I0	J2-27 SERIAL3_RX	
1	I1		
4	Y		38 PC7 UART6_SNIFF
6	SEL		21 PA5 UART_CH3-4
5	VCC 3V3_DBG		

U19 74LVC1G157GW

			DEBUG U15 STM32F405RGT6
3	I0	J1-34 SERIAL0_TX	
1	I1	J1-35 SERIAL1_TX	
4	Y		17 PA3 UART2_SNIFF
6	SEL		20 PA4 UART_CH1-2
5	VCC 3V3_DBG		

PB2 Pushbutton

6	3V3_DBG	60 DGB BOOT0

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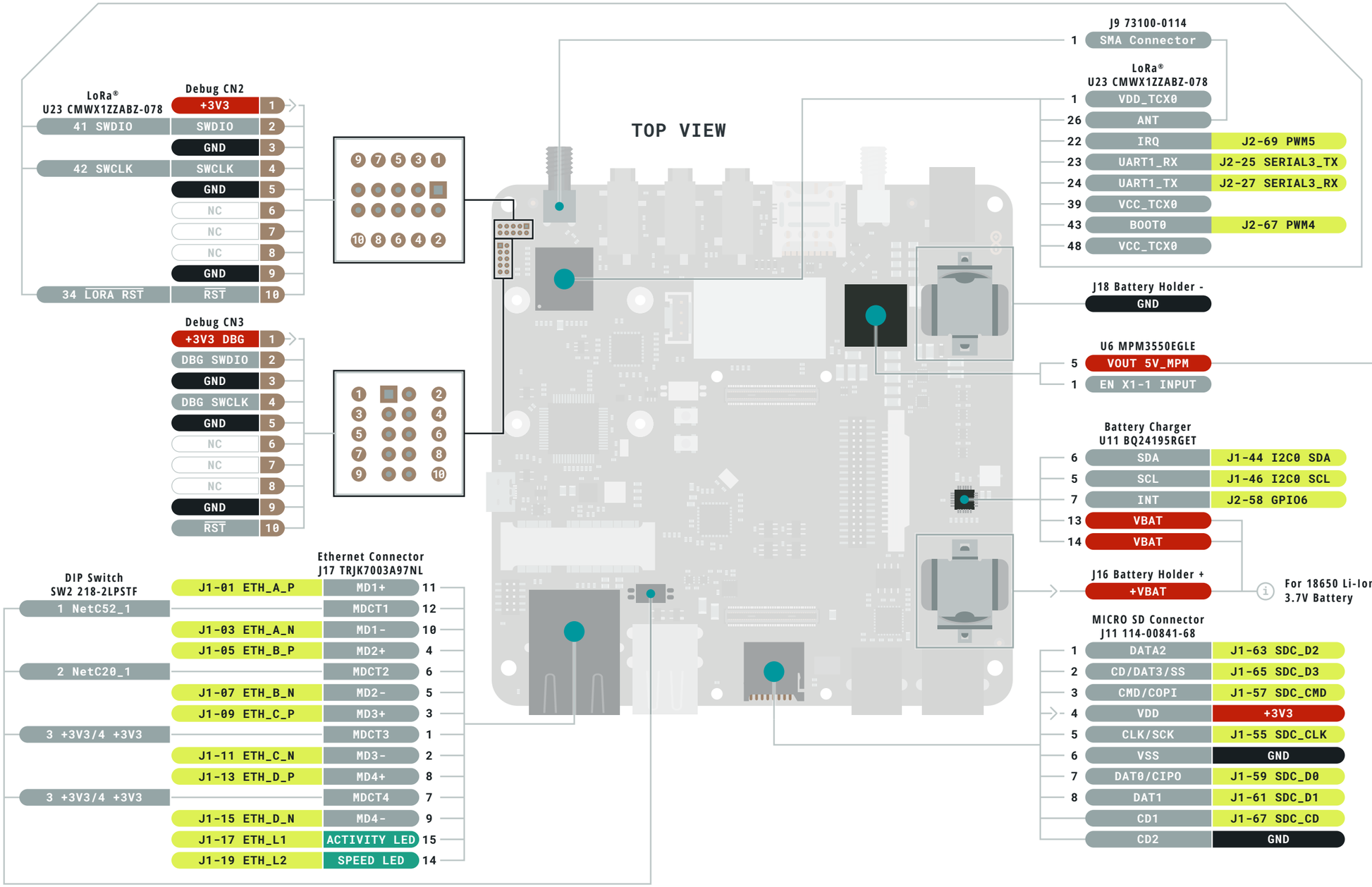
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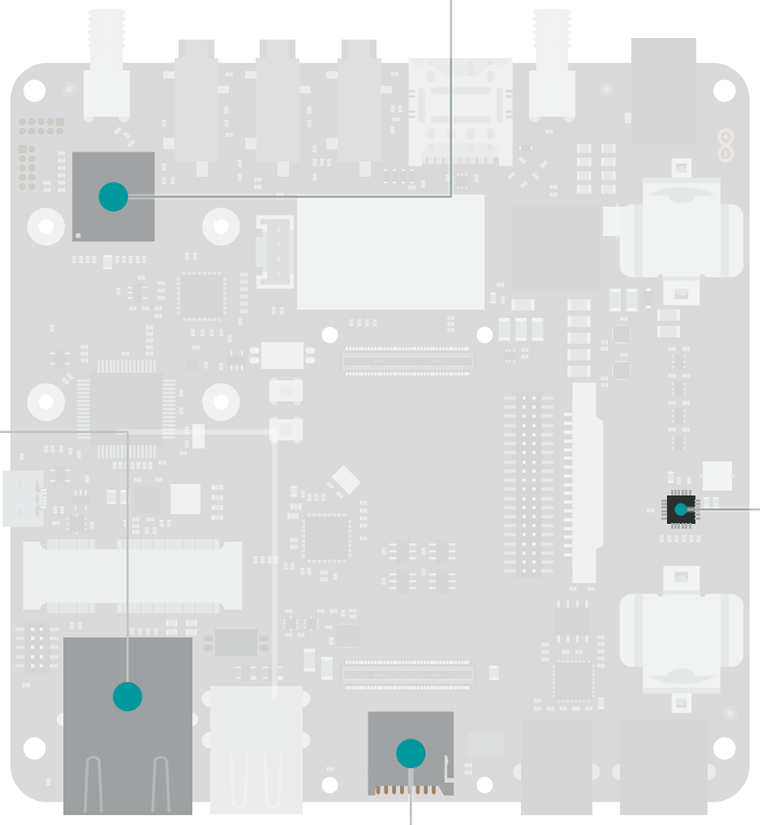
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TOP VIEW



LoRa®
U23 CMWX1ZZABZ-078

			GPIO (Linux)	Device Name (Linux)
1	VDD_TCX0			
26	ANT			
22	IRQ	J2-69 PWM5		
23	UART1_RX	J2-25 SERIAL3_TX	157	/dev/ttymxc3
24	UART1_TX	J2-27 SERIAL3_RX	156	/dev/ttymxc3
39	VCC_TCX0			
43	BOOT0	J2-67 PWM4	187	/sys/class/pwm/pwmchip0/pwm4
48	VCC_TCX0			

Ethernet Connector
J17 TRJK7003A97NL

Device Name (Linux)			
eth0 (network)	J1-01 ETH_A_P	MD1+	11
		MDCT1	12
eth0 (network)	J1-03 ETH_A_N	MD1-	10
	J1-05 ETH_B_P	MD2+	4
		MDCT2	6
eth0 (network)	J1-07 ETH_B_N	MD2-	5
	J1-09 ETH_C_P	MD3+	3
		MDCT3	1
eth0 (network)	J1-11 ETH_C_N	MD3-	2
	J1-13 ETH_D_P	MD4+	8
		MDCT4	7
eth0 (network)	J1-15 ETH_D_N	MD4-	9
	J1-17 ETH_L1	ACTIVITY LED	15
	J1-19 ETH_L2	SPEED LED	14

Battery Charger
U11 BQ24195RGET

			GPIO (Linux)	Device Name (Linux)
6	SDA	J1-44 I2C0 SDA		/dev/mmcblk1
5	SCL	J1-46 I2C0 SCL		/dev/mmcblk1
7	INT	J2-58 GPIO6	166	
13	VBAT			
14	VBAT			

MICRO SD Connector
J11 114-00841-68

			Device Name (Linux)
1	DATA2	J1-63 SDC_D2	/dev/mmcblk1
2	CD/DAT3/SS	J1-65 SDC_D3	/dev/mmcblk1
3	CMD/COPI	J1-57 SDC_CMD	/dev/mmcblk1
4	VDD	+3V3	
5	CLK/SCK	J1-55 SDC_CLK	/dev/mmcblk1
6	VSS	GND	
7	DAT0/CIPO	J1-59 SDC_D0	/dev/mmcblk1
8	DAT1	J1-61 SDC_D1	/dev/mmcblk1
	CD1	J1-67 SDC_CD	/dev/mmcblk1
	CD2	GND	

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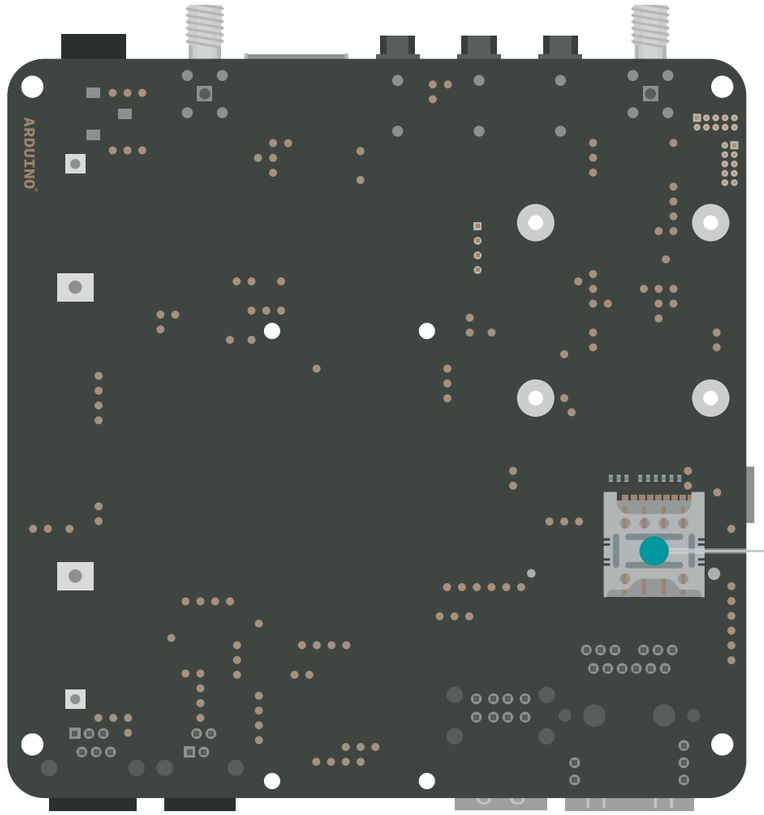
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BOTTOM VIEW



BUC Converter
U13 TPS54620RGYR

J2-62 PWM6	EN	10
	VIN 5V	6
Dedicated BUC Converter to provide 3V3_PCIE from 5V generated on board	PH 3V3_PCIe	11
	PH 3V3_PCIe	12

Mini PCIe Connector
J8 1759546-1

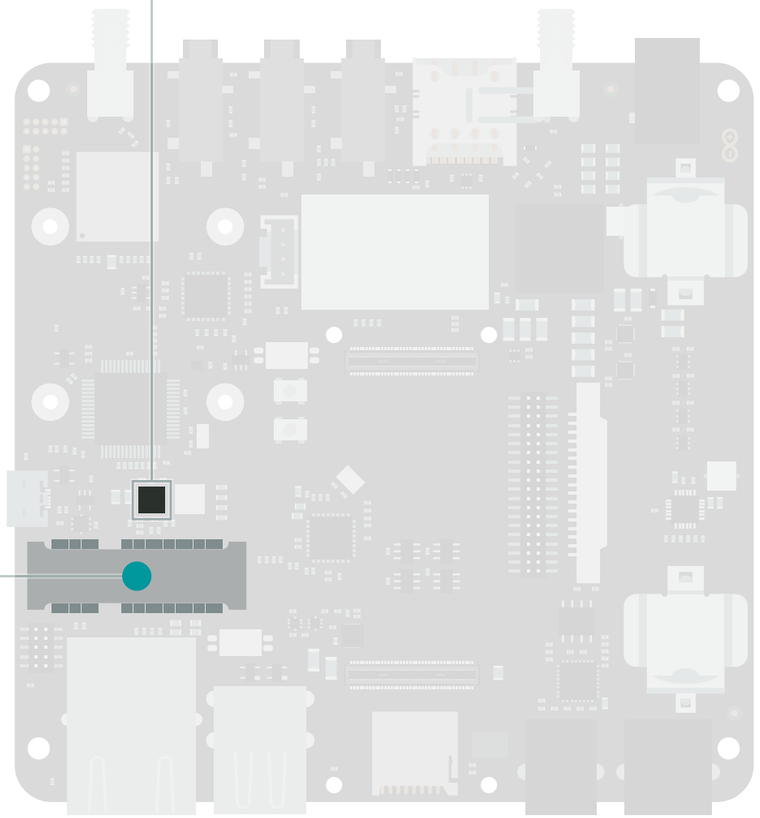
SIM2 Connector
SIM8060-6-0-14-00-A

	+3V3_PCIe	2
	GND	4
J2-68 PWM9	CLKREQ	7
C1 VCC	UIM_PWR	8
	GND	9
C7 I/0	UIM_DATA	10
J2-19 PCIe_CK_N	mPCIe_CK_N	11
C3 CLK	UIM_CLK	12
J2-17 PCIe_CK_P	mPCIe_CK_P	13
C2 RST	UIM_RST	14
	GND	15
	GND	18
J2-75 ANALOG_A1	W_DISABLE	20
	GND	21
J2-21 PCIe_RST	mPCIe_RST	22
J2-15 PCIe_RX_N	mPCIe_RX_N	23
	+3V3_PCIe	24
J2-13 PCIe_RX_P	mPCIe_RX_P	25
	GND	26
	GND	27
	GND	29
J1-46 I2C0 SCL	SCL	30
J2-11 PCIe_TX_N	mPCIe_TX_N	31
J1-44 I2C0 SDA	SDA	32
J2-09 PCIe_TX_P	mPCIe_TX_P	33
	GND	34
	GND	35

USB Hub
U3 USB2514B

8 USBDM_DN4	USB_DN	36
	GND	37
9 USBDP_DN4	USB_DP	38
	+3V3_PCIe	39
	GND	40
	+3V3_PCIe	41
	GND	43
	GND	50
	+3V3_PCIe	52

TOP VIEW



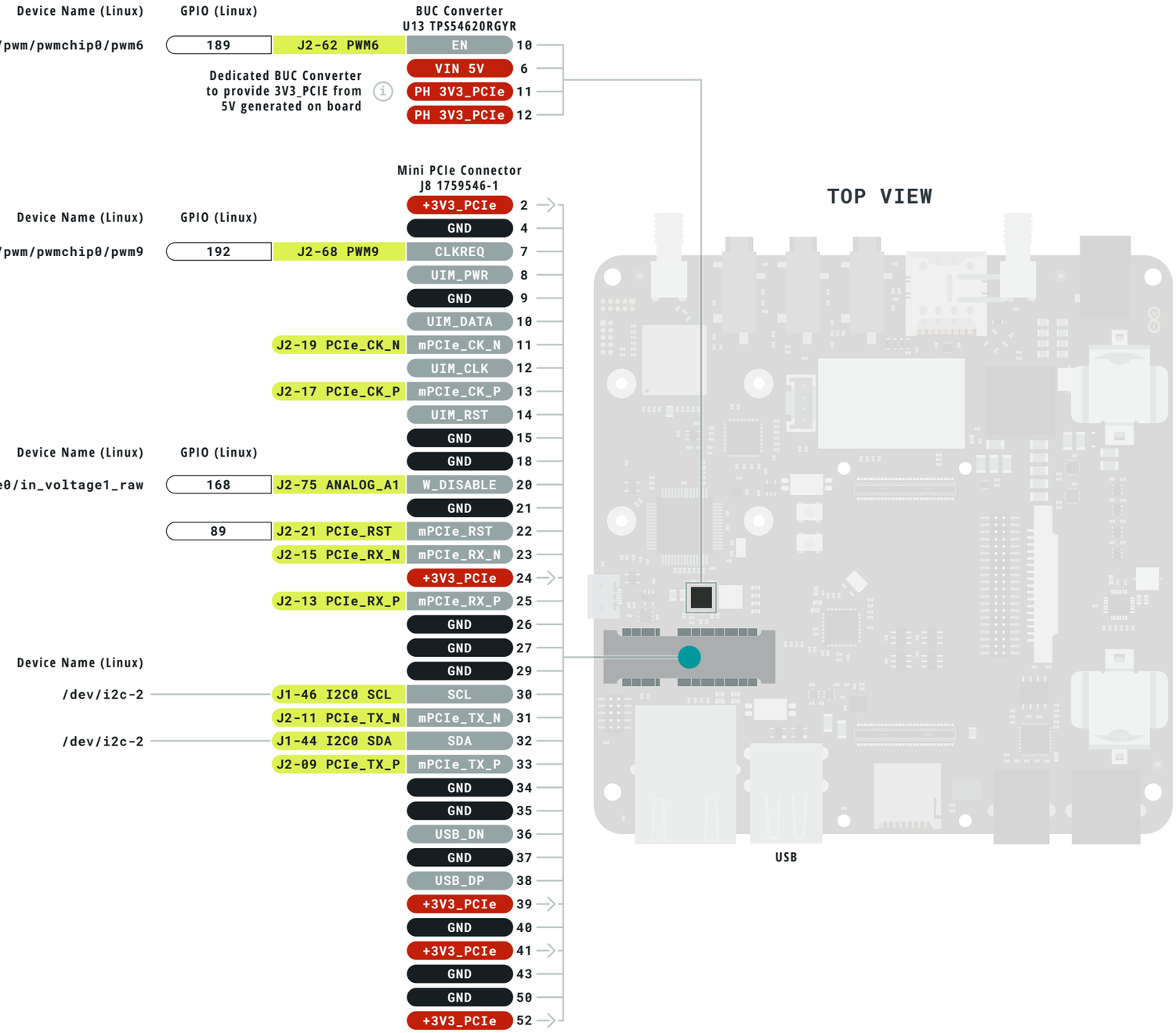
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Ground
Power
LED
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SWD Pin

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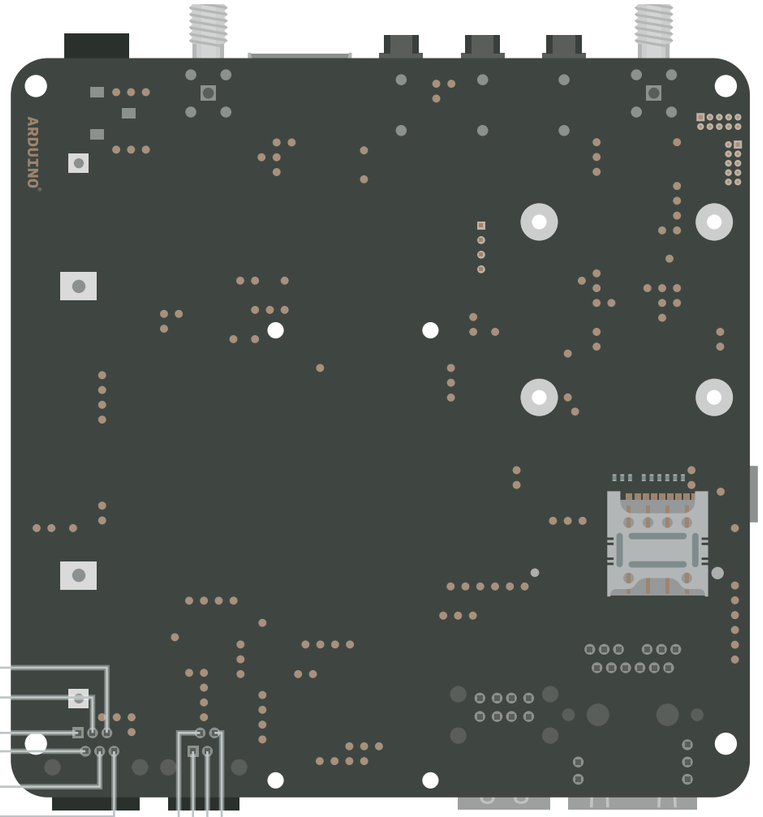
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ARDUINO PORTENTA MAX CARRIER

BOTTOM VIEW



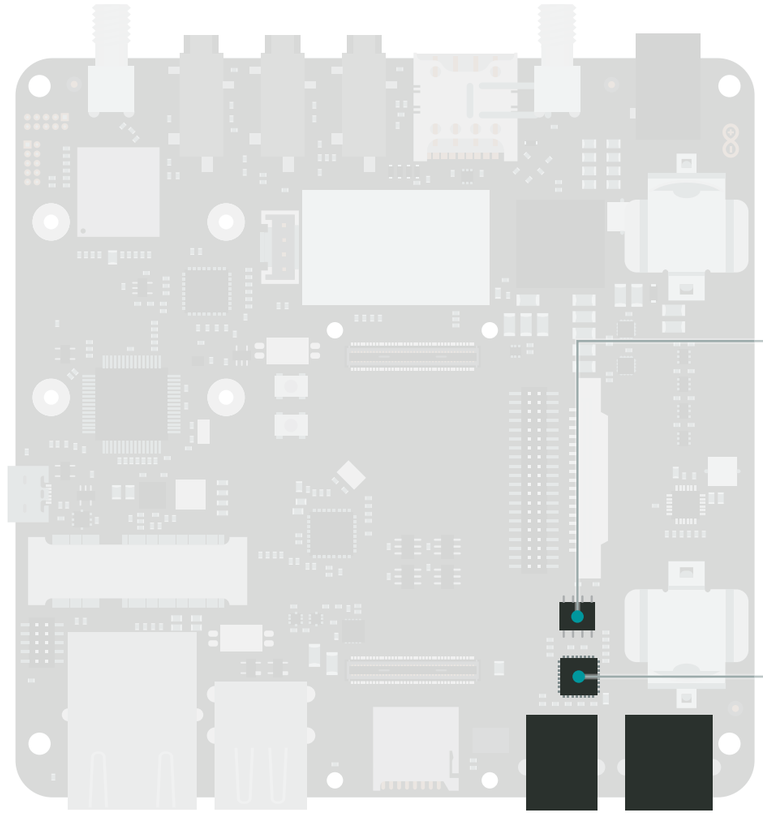
RS485/232 Connector
J6 615006138421

B	
Y	
GND	
Z	
A	
GND	

CAN Connector
J7 615004143421

CANH	
+5V	
CANL	
GND	

TOP VIEW



CAN
U5 TJA1049T/3J

1	TX	J1-49 CAN1 TX
2	RX	J1-51 CAN1 RX
6	CANL	
7	CANH	
8	STB	J1-65 PWM3

CAN Connector
J7 615004143421

3	CANL
2	CANH

RS485/232
U4 SP335ECR1-L

22	T1IN	J1-34 485_TX
21	T2IN	J1-38 485_RTS
20	$\overline{RE}$	J2-66 PWM8
7	R2OUT	J1-36 485_RX
6	R1OUT	J1-40 485_CTS
15	HALF/$\overline{FULL}$	J2-59 PWM0
14	RS-485/$\overline{RS-232}$	J2-61 PWM1
11	$\overline{SLEW}$	+3V3
12	FD_TX_TERM	J2-56 GPIO5
13	TERM	J2-64 PWM7
10	$\overline{SHDN}$	J2-63 PWM2

RS485/232 Connector
J6 615006138421

3	Z	2 Z
4	Y	3 Y
19	B	5 B
18	A	4 A

RS485/232 Connector
J6 615006138421

1	GND
2	Z
3	Y
4	A
5	B
6	GND

CAN Connector
J7 615004143421

1	+5V
2	CANH
3	CANL
4	GND

Protected with a 500mA PTC
(F1 0ZCG0050AF2C)

- Ground
- Power
- LED
- Internal Pin
- SWD Pin
- Digital Pin
- Analog Pin
- Other Pin
- High Density Connector

WARNING External power supply or battery is required to supply 5V which powers most of the peripherals

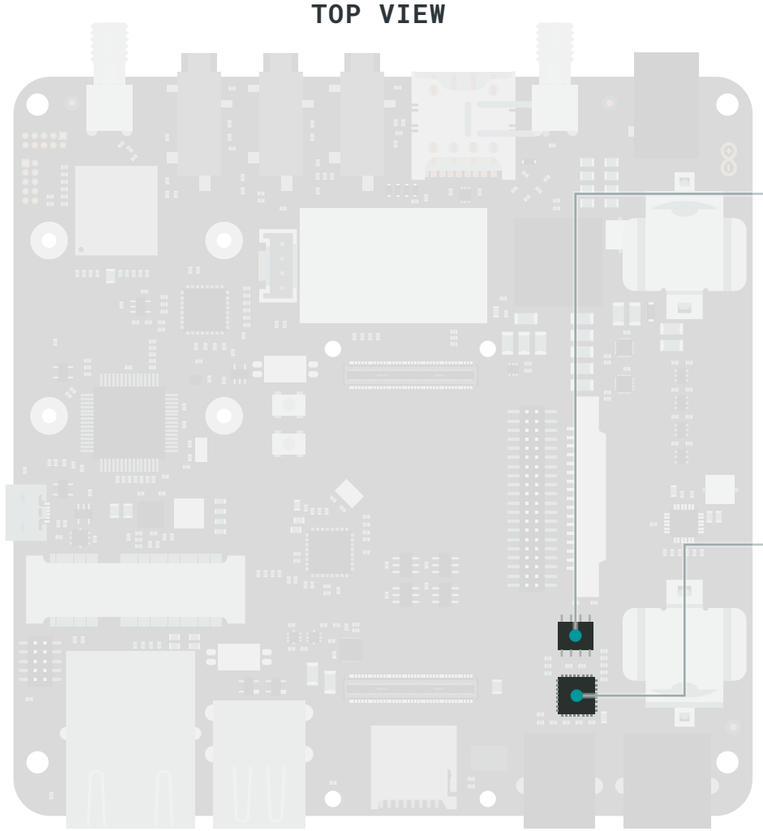
WARNING 3V3 voltage is generated by Portenta

ARDUINO.CC

Last update: 22 Oct, 2026

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TOP VIEW



			GPIO (Linux)	Device Name (Linux)
CAN U5 TJA1049T/3J				
1	TX	J1-49 CAN0 TX	177	can0 (network)
4	RX	J1-51 CAN0 RX	178	
6	CANL			
7	CANH			
8	STB	J1-65 PWM3		/dev/mmcblk1
RS485/232 U4 SP335ECR1-L				
22	T1IN	J1-34 485_TX	179	/dev/ttyX0
21	T2IN	J1-38 485_RTS	126	
20	$\overline{RE}$	J2-66 PWM8	191	/sys/class/pwm/pwmchip0/pwm8
7	R2OUT	J1-36 485_RX	127	/dev/ttyX0
6	R1OUT	J1-40 485_CTS	125	
15	HALF/$\overline{FULL}$	J2-59 PWM0	183	/sys/class/pwm/pwmchip0/pwm0
14	RS-485/$\overline{RS-232}$	J2-61 PWM1	184	/sys/class/pwm/pwmchip0/pwm1
11	$\overline{SLEW}$	+3V3		
12	FD_TX_TERM	J2-56 GPIO5	165	
13	TERM	J2-64 PWM7	190	/sys/class/pwm/pwmchip0/pwm7
10	$\overline{SHDN}$	J2-63 PWM2	185	/sys/class/pwm/pwmchip0/pwm2
3	Z			
4	Y			
19	B			
18	A			

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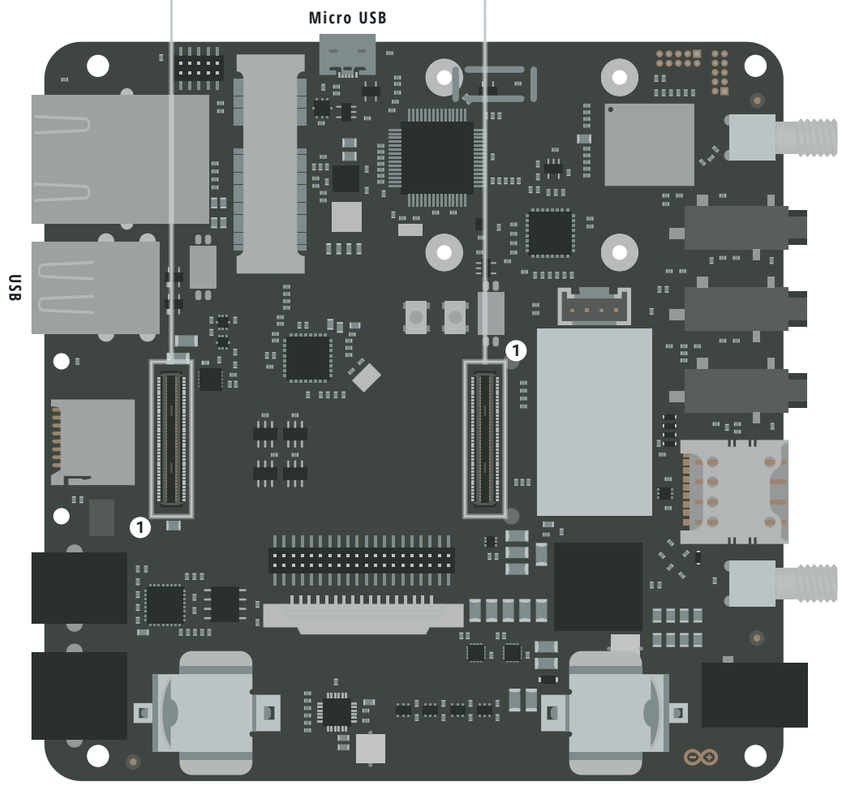
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ARDUINO

ARDUINO
PORTENTA MAX CARRIER

J1-Female

SWO	79	80	
SWCK	77	78	
SWDIO	75	76	
RESET	73	74	
	71	72	V-SDCARD
	69	70	
	67	68	DMIC D0
SDC D3	65	66	DMIC CK
SDC D2	63	64	VSYS
SDC D1	61	62	I2S DO
SDC D0	59	60	I2S DI
SDC CMD	57	58	I2S WS
SDC CLK	55	56	I2S MCK
VSYS	53	54	GND
CAN1 RX	51	52	
CAN1 TX	49	50	
GND	47	48	VIN
I2C1 SD6	45	46	I2C0 SCL
I2C1 SDA	43	44	I2C0 SDA
VIN	41	42	GND
UART1 CTS	39	40	UART0 CTS
UART1 RTS	37	38	UART0 RTS
UART1 RX	35	36	UART0 RX
UART1 TX	33	34	UART0 TX
GND	31	32	VIN
USB1 ID	29	30	
USB1 D-	27	28	USB0 D-
USB1 D+	25	26	USB0 D+
USB1 VBUS	23	24	
VIN	21	22	GND
ETH L2	19	20	DSI CK-
ETH L1	17	18	DSI CK+
	15	16	DSI D0-
	13	14	DSI D0+
	11	12	DSI D1-
	9	10	DSI D1+
ETH B-	7	8	
ETH B+	5	6	
ETH A-	3	4	
ETH A+	1	2	



TOP VIEW

J2-Female

CAM D7	2	1	FORCE_BOOTLOADER
CAM D6	4	3	BOOT_SOURCE
CAM D5	6	5	POWER_ON_REQ
CAM D4	8	7	COINCELL
CAM D3	10	9	
CAM D2	12	11	
CAM D1	14	13	
CAM D0	16	15	
CAM VS	18	17	
CAM CLK	20	19	
CAM HS	22	21	
GND	24	23	VCC/+3V3
UART2 TX	26	25	UART3 TX
UART2 RX	28	27	UART3 RX
	30	29	
	32	31	
VCC	34	33	GND
SPI1 CS	36	35	
SPI1 CK	38	37	
SPI1 CIPO	40	39	
SPI1 COPI	42	41	
GND	44	43	VCC/+3V3
GPIO 0	46	45	I2C2 SDA
GPIO 1	48	47	I2C2 SCL
GPIO 2	50	49	SAI CK
GPIO 3	52	51	SAI FS
GPIO 4	54	53	SAI D0
GPIO 5	56	55	
GPIO 6	58	57	GND
PWM 5	60	59	PWM 0
PWM 6	62	61	PWM 1
PWM 7	64	63	PWM 2
PWM 8	66	65	PWM 3
PWM 9	68	67	PWM 4
GND	70	69	VCC/+3V3
ADC VREF-	72	71	ADC VREF+
ADC A4	74	73	ADC A0
ADC A5	76	75	ADC A1
ADC A6	78	77	ADC A2
ADC A7	80	79	ADC A3

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